

CS439 Project Checkpoint 2 – Status Report

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Project Description

Portable devices such as wireless headphones, mice, and IoT sensors rely on Wi-Fi or Bluetooth/BLE for communication. Wi-Fi offers high throughput but consumes more energy, while Bluetooth/BLE provides lower energy usage but at reduced throughput. Current systems often stick to one communication interface/protocol and do not intelligently *combine* them. However, works such as [1] prove that there is potential for improved performance when using more than one interface/protocol at a time.

This project proposes to **leverage BLE to enhance Wi-Fi connection setups**. *In particular, the aim is to utilize BLE Advertisements as a Wake Control Mechanism for Wi-Fi connections, signaling existing Wi-Fi connections in sleep mode to wake up depending on observed data stream characteristics. Using relevant outgoing data telemetry – such as packet size, urgency, and type – an application-level algorithm will determine when BLE “wake up” advertisements should be sent to Wi-Fi endpoints in a manner that optimizes total power consumption.*

The goal of this project is to maximize throughput while minimizing energy consumption and improving the overall user experience for portable devices such as cellular phones, wireless mice, headphones, and IoT nodes. High power consumption and degraded user experience – particularly due to persistent network connectivity requirements and mobility-related challenges – have long been recognized issues. Prior work such as [2], [3], and [4] has explored these problems and proposed various solutions. Building on these efforts, we believe this project will contribute meaningfully to addressing these challenges.

Key Changes to Project Approach

Since the Extended Project Proposal (Checkpoint #1), much of our approach to the project has changed. *Most importantly, after collecting precise power/energy consumption data for both the Wi-Fi and BLE protocols, we have determined that our initial plan of “smart handover” between the two protocols will not lead to optimized power consumption and throughput*, and have thus shifted our approach to a BLE Advertisement-based Wake Control Algorithm instead.

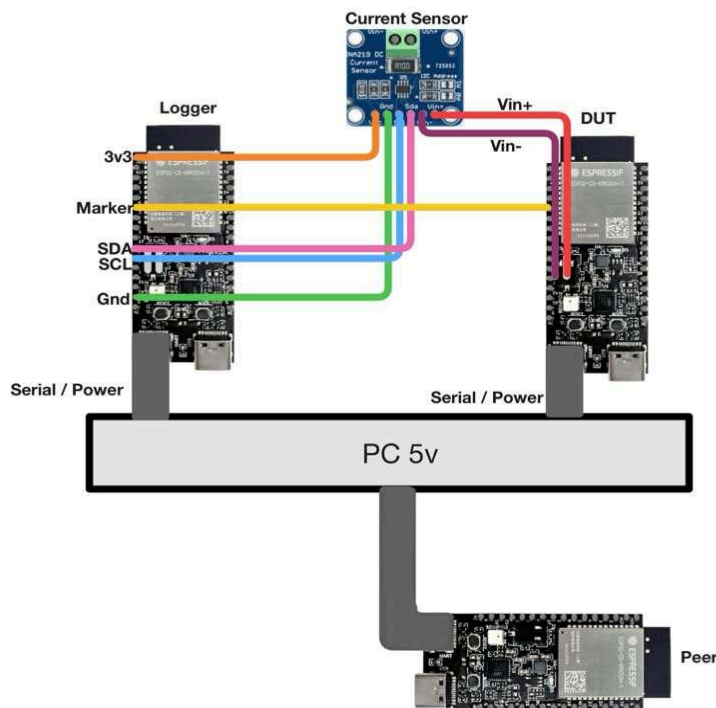
To inform our approach, **we collected power measurements using an Adafruit INA219 DC Current Sensor Breakout Board**. Our setup and methodology for this data collection is described in the “**Data Collection**” section below.

Data Collection

To determine the best approach, we collected real-time power, energy, throughput, and total transfer time statistics for both the Wi-Fi 6 (802.11ax) and BLE 5 protocols, running between 2 ESP32-C5 microcontrollers. One ESP, labeled as the “**Device Under Test**” (DUT) below, initiates data transfers of varying payload sizes to another ESP, labeled as the “**Peer**”. A third ESP, labeled as the “**Logger**”, is used to offload data collection and logging from the ESPs involved in the data transfers to ensure logging does not interfere with the collected data itself.

An **Adafruit INA219 DC Current Sensor Breakout Board** collects precise power measurements from the DUT during data transfers, and its measurements are offloaded to and processed by the Logger. Finally, for real-time signaling to the Logger for data collection, the DUT leverages a GPIO pin, labeled as the “**Marker**” (yellow wire).

This setup can be seen in the pin-out diagram below:



For comprehensive testing, we ran 21 iterations of data collection:

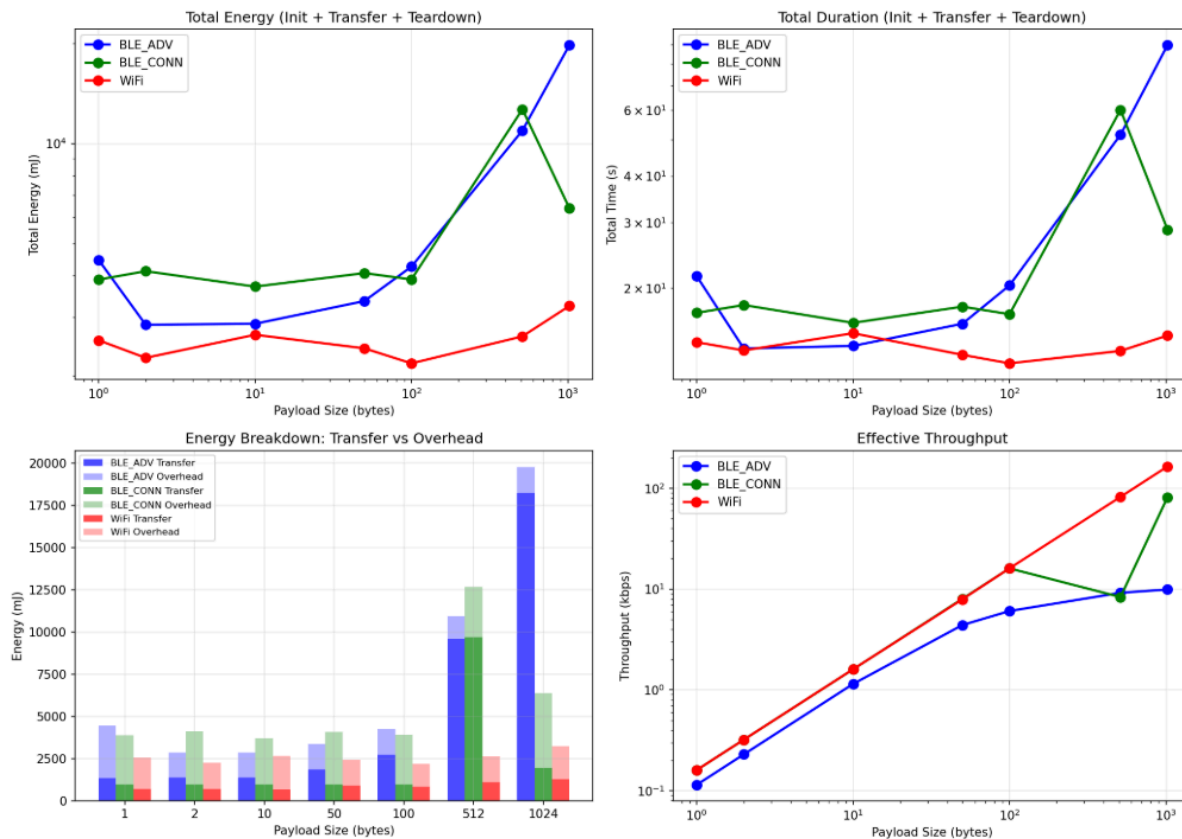
- 7 different payload sizes (in bytes): 1, 2, 10, 50, 100, 512, 1024
- 3 different connection modes for data transfers: Wi-Fi, BLE Advertisements (BLE_ADV), BLE Connected Mode (BLE_CONN)

Our results are displayed and analyzed in the “**Data Analysis**” section below.

Data Analysis

Our data collection is broken down into two categories: **Full Energy**, which includes the entire data transfer from start to finish (including connection setup and teardown), and **Transfer Phase Only**. They are each shown and analyzed below:

Full Energy (Init + Transfer + Teardown)



From this data, the following conclusions arise:

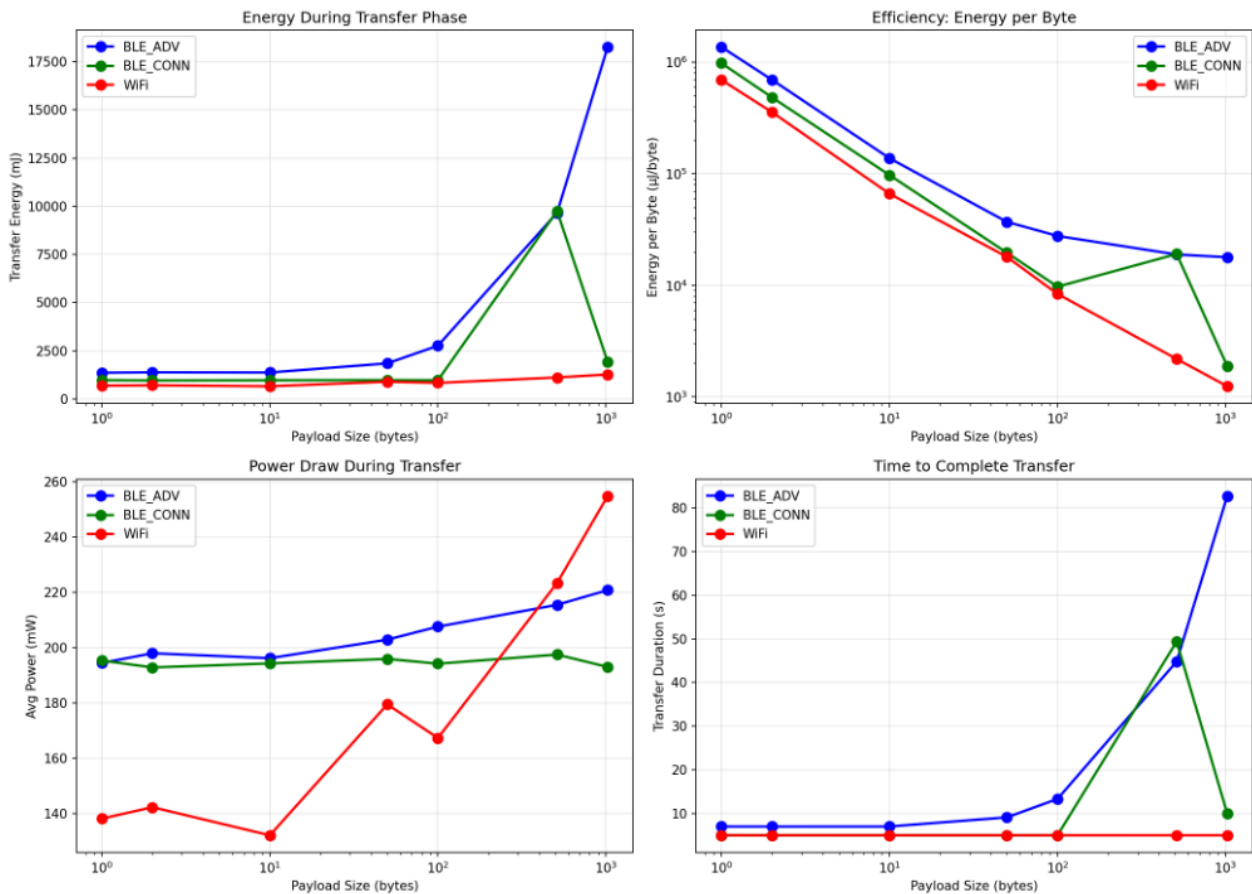
1. **BLE is inferior for large transfers:**

- As the payload size increases, BLE energy consumption, total transfer time, overhead cost, and throughput are significantly worse than Wi-Fi for both the advertising and connection BLE modes!

2. **BLE Advertisements incur the lowest relative setup energy overhead:**

- As evident in the bottom left chart, both Wi-Fi and BLE's connection mode incur more overhead energy consumption than consumption during the transfer itself. Only BLE Advertisements consistently have less energy consumption during the "overhead" phase.
- Since we define the overhead phase here as the connection setup + teardown, this means **BLE advertisements are the ideal tool for a lightweight wake-up mechanism!**

Transfer Phase Only



From this data, the following conclusions arise:

3. BLE is inferior for large transfers (again):

- In addition to the previously mentioned results, BLE modes also experience the highest energy consumption per byte (top right graph) and higher transfer times and energy consumption for transfers (top left/bottom right graphs, respectively)

4. Wi-Fi is power hungry:

- Despite being superior for data transfers, Wi-Fi experiences a rapidly increasing power draw as you increase the payload. (bottom left graph)

Data Analysis Conclusions: BLE should NOT be used for bulk data transfers in ANY capacity. As concluded by #1 and 3, BLE modes are way more inefficient at bulk data transfer to be used regularly over Wi-Fi, no matter the circumstance. However, if BLE were to be used to optimize something, it would be to optimize power draw (#4: Wi-Fi is power hungry), and it could best be done by exploiting BLE Advertisements' lower setup energy consumption (#2).

It is due to these conclusions that we have shifted our approach from “smart handover” to “BLE Advertisement-based Wi-Fi Wake-Up signaling”.

Conclusion

With this data collection setup and analysis complete, we are now well-equipped to effectively complete this project. It is now clear that our algorithm will focus on leveraging BLE advertisements to signal Wi-Fi connections in sleep mode to wake up. One basic approach to this algorithm is as follows, for the scenario of video streaming:

Receiver Side:

1. Play video from buffer
2. When buffer < threshold :
 - a. Start BLE advertising "NEED_DATA"
3. When WiFi data arrives:
 - a. Stop BLE advertising
 - b. Buffer the data

Sender Side:

1. Deep sleep (very low power)
2. Periodic BLE scan (~100ms every few seconds)
3. If "NEED_DATA" advertisement detected:
 - a. Wake up Wi-Fi
 - b. Send a burst of video data
 - c. Shut down Wi-Fi
 - d. Return to sleep

We will also look into using other metrics besides binary thresholds to inform Wake-up more effectively. Finally, we will also use more state-of-the-art metrics, such as Wi-Fi Duty Cycle estimation, to back our algorithm's performance.

References

- [1] [K. Chebrolu and A. Rao. Bandwidth aggregation for real-time applications in heterogeneous wireless networks. IEEE Transactions on Mobile Computing, 2006.](#)
- [2] [A. Rahmati and L. Zhong. Context-for-wireless: Context-sensitive energy-efficient wireless data transfer. MobiSys 2007.](#)
- [3] [M. Kassar, B. Kervella, and G. Pujolle. An overview of vertical handover decision strategies in heterogeneous wireless networks. Computer Communications, 2008.](#)
- [4] [S. Yin, S. Zhang, and O. Li. Energy-efficient wireless communication for IoT: A survey. IEEE Communications Surveys & Tutorials, 2020.](#)